

Neurological studies have demonstrated that fear-inducing experiences are rapidly stored as memories by neurons and neuronal connections in various brain regions, including the hippocampus and medial prefrontal cortex (mPFC). During cellular activity involved in fear-induced memory storage, certain locations in the genomes of brain neurons and astrocytes undergo double-stranded DNA breaks (DSBs). These DSBs modulate DNA packing to allow cellular machinery better access to genes involved in memory storage processes.

Fear-inducing experiences also promote adrenal glucocorticoid release. Glucocorticoid receptors (GR), which act as transcription factors, are located in the cytosol of cells throughout the body and are activated by glucocorticoid binding.

To test fear-induced neurological effects, researchers performed contextual fear conditioning (CFC) in mice. During CFC, mice were exposed to an electric shock upon entering a chamber, resulting in fear memory generation. One group of mice (CFC-30 mice) underwent CFC whereas another age-matched control group (naïve mice) did not undergo CFC. Thirty minutes after CFC, researchers measured the relative expression levels of two genes in the hippocampus and mPFC of both the CFC-30 mice and naïve mice (Figure 1). These genes, *Npas4* and *Arc*, are involved in memory formation and storage.

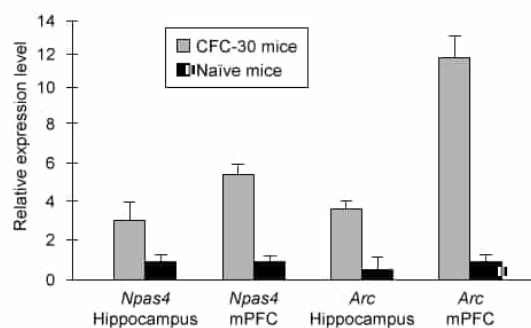


Figure 1 Relative expression levels of *Npas4* and *Arc* in CFC-30 and naïve mice in the hippocampus and mPFC

Researchers next examined whether *Sgk1*, a gene regulated by glucocorticoids, is dependent on GR expression to induce a response to CFC. A group of mice was treated with RU-486, a GR antagonist, and then underwent CFC. Expression of *Sgk1* was measured 30 minutes after CFC in these mice (CFC-30 + RU-486) as well as in mice that underwent CFC without prior RU-486 treatment (CFC-30). *Sgk1* expression was also measured in mice that did not undergo CFC or RU-486 treatment (naïve) (Figure 2).



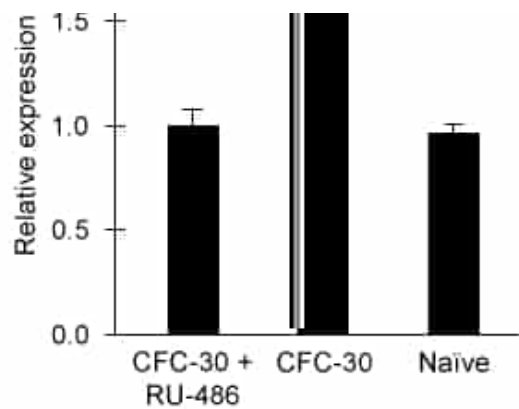


Figure 2 Relative expression levels of *Sgk1* in CFC-30 + RU-486, CFC-30, and naïve mice

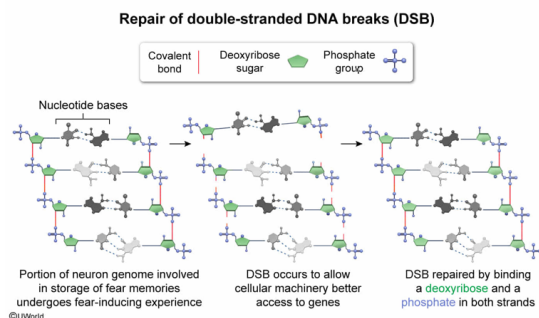
DSBs that occur to modulate DNA packing after a fear-inducing experience are generally repaired. These repairs require binding between:

- ☐ A. a purine and a ribose. [1%]
- ☐ B. a deoxyribose and a pyrimidine. [15%]
- ✓ ☒ C. a deoxyribose and a phosphate. [73%]
- ☐ D. a phosphate and a ribose. [8%]

Correct

73% answered correctly

Explanation:



DNA is a long, double-stranded polymer made of **deoxyribonucleotide monomers** and consists of an exterior "backbone" and an interior "ladder." The two strands are aligned in an **antiparallel direction** and wrap around one another to form a double helix.

The exterior backbone of DNA consists of deoxyribose (5-carbon/pentose) sugars and phosphate groups that alternate with one another. The 3'-hydroxy group of the deoxyribose from a deoxyribonucleotide monomer binds via **phosphodiester linkage** to the 5'-phosphate group of another deoxyribonucleotide monomer.

The interior ladder of DNA is composed of complementary **nitrogenous bases** attached to the deoxyribose sugars on the backbone of both strands. DNA contains four nitrogenous bases: the purines adenine (A) and guanine (G), and the pyrimidines cytosine (C) and thymine (T). These bases are held together in this ladder formation via hydrogen bonds.

The passage states that during cellular activity involved in fear-induced memory storage, certain

locations in the **genomes** of brain **neurons** and astrocytes undergo double-stranded DNA breaks (DSBs). This modulates **DNA packing** to allow cellular machinery (eg, **RNA polymerase**) better access to the genes involved in memory storage processes. Because these breaks are double-stranded, they must occur in the backbone, between adjacent phosphate groups and deoxyribose sugars.

As given in this question, these DSBs are generally repaired after they occur. To repair DSBs, a **deoxyribose** sugar and a **phosphate** on both DNA strands must bind (ie, via phosphodiester bonds). This binding rejoins the breaks in the DNA double helix.

(Choices A and D) **Ribose sugars** are found in **RNA**, not DNA. Therefore, repair of DSBs would *not* require binding of a purine (A, G) or a phosphate to a *ribose*.

(Choice B) Deoxyribose sugars and pyrimidines (C, T) bind to form nucleosides. However, DSBs occur between the deoxyribose sugar of one nucleotide and the phosphate group of another nucleotide. Therefore, binding between a deoxyribose and a pyrimidine is *not* required to repair DSBs.

Educational objective:

DNA is a double-stranded polymer made of nucleotides. The backbone of DNA consists of deoxyribose sugars and phosphate groups that alternate with one another. Projecting from this backbone are four nitrogenous bases (adenine, thymine, guanine, and cytosine) that bind complementarily with the corresponding nitrogenous base in the second, antiparallel DNA strand.

Time spent:0 sec
Subject:
Biology

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System:
Molecular Biology

Neurological studies have demonstrated that fear-inducing experiences are rapidly stored as memories by neurons and neuronal connections in various brain regions, including the hippocampus and medial prefrontal cortex (mPFC). During cellular activity involved in fear-induced memory storage, certain locations in the genomes of brain neurons and astrocytes undergo double-stranded DNA breaks (DSBs). These DSBs modulate DNA packing to allow cellular machinery better access to genes involved in memory storage processes.

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To test fear-induced neurological effects, researchers performed contextual fear conditioning (CFC) in mice. During CFC, mice were exposed to an electric shock upon entering a chamber, resulting in fear memory generation. One group of mice (CFC-30 mice) underwent CFC whereas another age-matched control group (naïve mice) did not undergo CFC. Thirty minutes after CFC, researchers measured the relative expression levels of two genes in the hippocampus and mPFC of both the CFC-30 mice and naïve mice (Figure 1). These genes, *Npas4* and *Arc*, are involved in memory formation and storage.

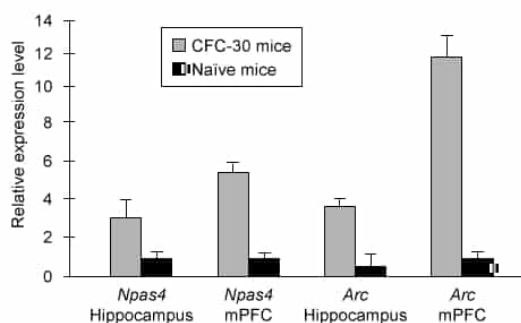
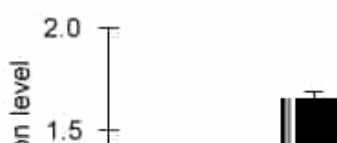


Figure 1 Relative expression levels of *Npas4* and *Arc* in CFC-30 and naïve mice in the hippocampus and mPFC

Researchers next examined whether *Sgk1*, a gene regulated by glucocorticoids, is dependent on GR expression to induce a response to CFC. A group of mice was treated with RU-486, a GR antagonist, and then underwent CFC. Expression of *Sgk1* was measured 30 minutes after CFC in these mice (CFC-30 + RU-486) as well as in mice that underwent CFC without prior RU-486 treatment (CFC-30). *Sgk1* expression was also measured in mice that did not undergo CFC or RU-486 treatment (naïve) (Figure 2).



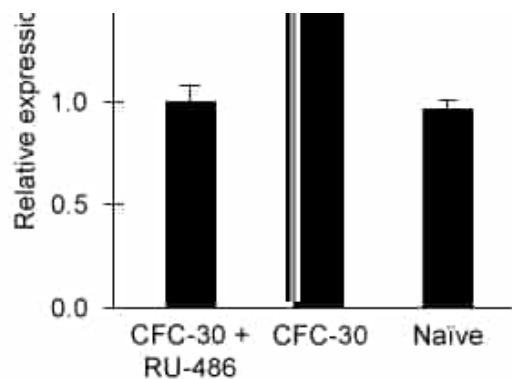


Figure 2 Relative expression levels of *Sgk1* in CFC-30 + RU-486, CFC-30, and naïve mice

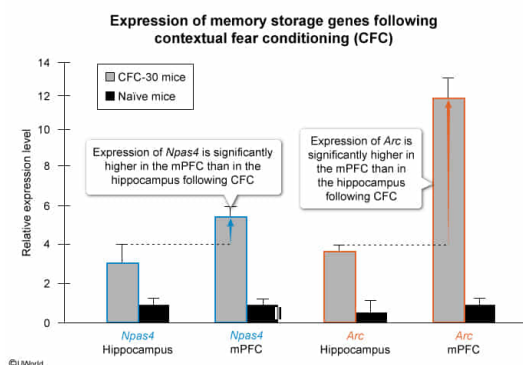
Which statement regarding gene expression is best supported by the data in Figure 1?

- ☐ A. CFC-30 mice express significantly more hippocampal *Arc* than hippocampal *Npas4*. [5%]
- ☒ B. Expression of memory storage genes is higher in the mPFC than in the hippocampus following CFC. [79%]
- ☐ C. Naïve mice differ significantly in the expression of *Npas4* and *Arc* in the mPFC. [8%]
- ☐ D. CFC does not affect expression of memory storage genes in naïve mice. [6%]

Correct

78% answered correctly

Explanation:



Gene expression is controlled so that the appropriate genes are expressed in the correct amounts, in the right cells, and at the proper times. Although several mechanisms exist for controlling gene expression, the primary mechanism is at the **transcriptional** level.

In the passage, researchers investigated the neurological effects of fear in mice. To do so, they performed contextual fear conditioning (CFC), in which mice were exposed to an electric shock

upon entering a chamber, resulting in the creation of a fear memory. As a control, naïve mice not undergoing CFC were also examined.

Figure 1 shows the relative expression levels of two genes (*Npas4* and *Arc*) involved in memory formation and storage in the hippocampus and mPFC of mice 30 minutes after undergoing contextual fear conditioning (CFC-30 mice). The relative expression levels of these two genes were also measured in mice not undergoing CFC (naïve mice) in the same brain regions. The

graph demonstrates that expression of both genes is **significantly higher** (ie, no error bar overlap) **in the mPFC than in the hippocampus following CFC**. Therefore, the mPFC is likely more highly involved in forming and storing fear memories than the hippocampus.

(Choice A) No significant difference is found in expression levels of hippocampal *Arc* and hippocampal *Npas4* in CFC-30 mice.

(Choice C) Naïve mice do not differ significantly in expression levels of *Npas4* and *Arc* in the mPFC.

(Choice D) According to the passage, naïve mice did not undergo CFC. Therefore, it *cannot* be determined whether CFC affects expression of memory storage genes in naïve mice.

Educational objective:

Gene expression is controlled at several levels to ensure that each cell type in an organism expresses the correct type and amount of corresponding protein at the right time.

Time spent:0 sec

Subject:
Biology

QID:403469

System:
Molecular Biology

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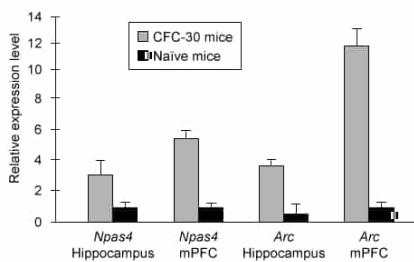


Figure 1 Relative expression levels of *Npas4* and *Arc* in CFC-30 and naïve mice in the hippocampus and mPFC

Researchers next examined whether *Sgk1*, a gene regulated by glucocorticoids, is dependent on GR expression to induce a response to CFC. A group of mice was treated with RU-486, a GR antagonist, and then underwent CFC. Expression of *Sgk1* was measured 30 minutes after CFC in these mice (CFC-30 + RU-486) as well as in mice that underwent CFC without prior RU-486 treatment (CFC-30). *Sgk1* expression was also measured in mice that did not undergo CFC or RU-486 treatment (naïve) (Figure 2).

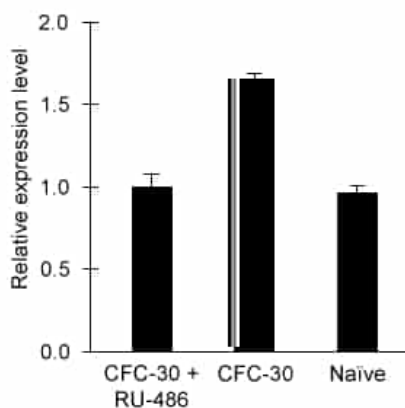


Figure 2 Relative expression levels of *Sgk1* in CFC-30 + RU-486, CFC-30, and naïve mice

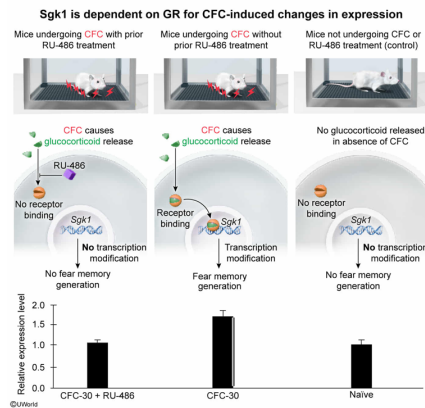
Do the data presented in Figure 2 support the hypothesis that *Sgk1* is dependent on GR for CFC-induced changes in expression?

- ✓ ☒ A. Yes; RU-486 treatment reduced expression of *Sgk1* to control levels in CFC-30 mice. [74%]
- ☐ B. Yes; expression of *Sgk1* increased in the presence of a GR antagonist in CFC-30 mice. [3%]
- ☐ C. No; naïve mice did not differ from CFC-30 mice treated with RU-486 in *Sgk1* expression, suggesting that *Sgk1* is not dependent on GR for CFC-induced expression changes. [17%]
- ☐ D. No; *Sgk1* expression did not significantly decrease in mice experiencing CFC after treatment with a GR antagonist, suggesting that GR does not affect *Sgk1* expression. [3%]

Correct

75% answered correctly

Explanation:



Transcription factors are proteins that alter **gene expression** through binding DNA upstream from promoter regions. Accordingly, transcription factors can either increase transcription (ie, **activators**) or decrease transcription (ie, repressors) of a specific gene. For example, glucocorticoids **released** from the adrenal gland bind transcription factors known as glucocorticoid receptors (GR). These transcription factors are located in the cytosol of cells throughout the body and influence the expression of genes following stimulation (eg, after a fear-inducing experience).

In the passage, researchers examined whether *Sgk1*, a gene regulated by glucocorticoids, is dependent on GR expression to induce a response to CFC. To determine this dependency, the researchers treated a group of mice with RU-486, a GR antagonist, and then performed CFC on these mice by exposing them to an electric shock upon entering a chamber. Thirty minutes after CFC, *Sgk1* expression was measured in these mice (CFC-30 + RU-486). The researchers also measured *Sgk1* expression in mice undergoing CFC without prior RU-486 treatment (CFC-30) and in mice that did not undergo CFC or RU-486 treatment (naïve).

Figure 2 shows that CFC performed in the absence of RU-486 (ie, CFC-30 mice) induces significantly increased expression of *Sgk1* compared to control (ie, naïve) mice. However, the data also show that prior treatment with the GR

antagonist RU-486 (ie, CFC-30 + RU-486 mice) reduced the expression of *Sgk1* to control levels following CFC.

Therefore, **the hypothesis** that *Sgk1* is **dependent** on GR for CFC-induced changes in expression **is supported by these data** because in the **absence of GR**, *Sgk1* expression **significantly decreased**.

(Choice B) Expression of *Sgk1* **decreased** (not increased) in the presence of a GR antagonist (ie, RU-486) in mice undergoing CFC (ie, CFC-30 + RU-486 mice).

(Choice C) Figure 2 shows that naïve mice did **not significantly differ** from CFC-30 mice treated with RU-486 in *Sgk1* expression. However, this result supports (not refutes) the hypothesis that *Sgk1* is dependent on GR for CFC-induced changes in expression because in the absence of GR, *Sgk1* expression following CFC **decreased**.

(Choice D) Expression of *Sgk1* **did significantly decrease** in mice experiencing CFC after treatment with a GR antagonist (ie, CFC-30 + RU-486 mice) compared to mice undergoing CFC without the GR antagonist (ie, CFC-30 mice). Therefore, the data suggest that GR **does** affect *Sgk1* expression.

Educational objective:

Transcription factors are proteins that change gene expression through binding DNA near promoter regions. This binding serves to either increase or decrease transcription of a specific gene.

Time spent: 0 sec

Subject:
Biology

QID: 403470

System:
Molecular Biology

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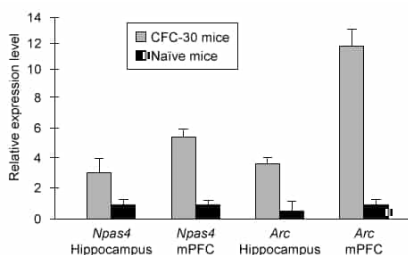


Figure 1 Relative expression levels of *Npas4* and *Arc* in CFC-30 and naïve mice in the hippocampus and mPFC

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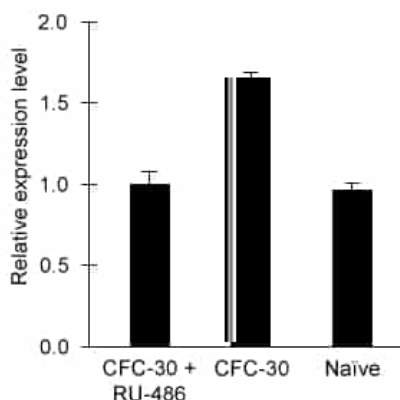


Figure 2 Relative expression levels of *Sgk1* in CFC-30 + RU-486, CFC-30, and naïve mice

cDNA was generated and amplified via PCR for several genes involved in memory formation. The following primer sequence was used to generate the cDNA for *Npas4*:

5'-TTTTAAGGCGATCAGCATCCAGAG-3'

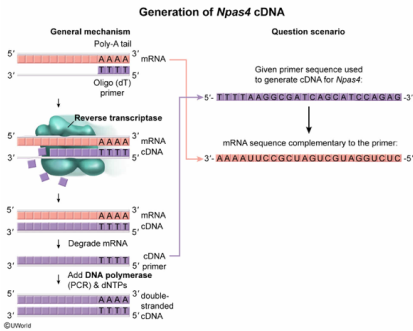
Which of the following nucleotide sequences represents the molecule from which this cDNA was generated?

- ☐ A. 3'-AAAATTCGCTAGTCGTAGGTCTC-5' [50%]
- ☐ B. 3'-CTCTGGATGCTGATCGCCTTAAAA-5' [8%]
- ☒ C. 3'-AAAAUCCGCUAGUCGUAGGUCUC-5' [37%]
- ☐ D. 3'-CUCUGGAUGCUGAUCGCCUAAAA-5' [4%]

Correct

37% answered correctly

Explanation:



When scientists are interested in specific mRNA transcripts (ie, expressed genes), reverse transcriptase polymerase chain reaction (RT-PCR) can be used to analyze cells. Reverse transcriptase (an enzyme) is used to convert mRNA to double-stranded cDNA, which can then be amplified through PCR using primers specific to the mRNA transcript of interest. Afterward, samples can be analyzed using gel electrophoresis for the presence or absence of a specific transcript.

In this question, cDNA was generated and amplified via PCR for several genes involved in memory formation. The primer sequence used to generate the cDNA for *Npas4* is given. To determine the nucleotide sequence of the molecule from which this cDNA was generated, the mRNA strand complementary to the given cDNA primer must be found. Using base-pairing rules and proper antiparallel strand orientation, the correct mRNA strand is represented by the sequence shown in Choice C.

(Choice A) Because this sequence contains thymine rather than uracil, it represents the correctly oriented DNA (not mRNA) sequence complementary to the cDNA primer.

(Choice B) This sequence represents the incorrect orientation of the DNA (not mRNA) sequence complementary to the cDNA primer.

(Choice D) This sequence represents the correct mRNA nucleotide sequence complementary to the cDNA primer but in the incorrect orientation.

Educational objective:

cDNA can be generated by converting mRNA with reverse transcriptase. The cDNA then undergoes reverse transcriptase polymerase chain reaction (RT-PCR) using primers specific to an mRNA transcript of interest for analysis.

Time spent:0 sec
Subject:
Biology

QID:403471
System:
Molecular Biology